

Aniruddha PANDEY

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This resume is a small language model! (Well, a tiny one, ~9M parameters) Run it on self-aware-resume.vercel.app. After 3+ years as a senior frontend engineer shipping production systems, I returned to school to focus on building intelligent systems from first principles. I'm now a 3rd-semester master's student at Stevens Institute of Technology. I'm fortunate to be advised by Dr. Nikhil Muralidhar. My current research interests are emphasis on mechanistic interpretability and agent memory architectures.

EDUCATION

- DEC 2026 (EXPECTED) **M.S. in Computer Science** - STEVENS INSTITUTE OF TECHNOLOGY, HOBOKEN, NJ
Coursework: Statistical Machine Learning, Deep Learning, NLP, Scientific Discovery in ML, Advanced Algorithms; Volunteer organizer, Next-Gen Healthcare Innovators Symposium 2025 — managed panel curation, speaker coordination, event logistics, and student research pitch presentations.
- SEP 2021 **B.Tech in Computer Science and Engineering** - LOVELY PROFESSIONAL UNIVERSITY, INDIA

GRADUATE RESEARCH

- JAN 2025 PRESENT **Behavioral Fingerprint Detection:** Framework exploiting divergence tokens—sub-semantic artifacts from LLM generation—for model-family attribution across **GPT-4o**, **Claude**, **Gemini**, **Llama**, and **Mistral**; multi-probe pipeline targeting AUROC ≥ 0.90 against humanization attacks. *(In preparation)*
- Tri-Mem:** Modality-matched memory hierarchies for multi-agent systems; three-layer cognitive architecture mapping semantic memory (sparse-attention KV chunks), episodic memory (visual compression), and declarative memory (vector RAG) with entropy-driven routing. Extends agent memory horizon from ~30 to 100+ turns at 90% token reduction; evaluated on a custom NovaCorp AuditBench with five agent ablations. *(In preparation)*
- NeuralSINDy:** Extending **SINDy** with neural function approximators to discover governing equations from noisy, high-dimensional dynamical systems data. *(In preparation)*

EXPERIMENTS

resume-llm: **ELF/PDF** polyglot containing a 3.1M-parameter ternary transformer with weights in $\{-1, 0, 1\}$; built the full pipeline spanning **BPE** tokenization, two-stage training (**SimpleStories** pretrain $\rightarrow$ document finetune) on $4 \times \mathbf{H100}$ (80GB) on Stevens' **Jarvis** cluster, packed binary export (~786KB), a **C** inference engine with ternary matrix multiplication, and **ELF/PDF** header interleaving via **.note** section injection.

Wikipedia Search Engine: Full-text search engine in **Rust** with a **BFS** crawler, inverted index using delta and **VByte** encoding (75% compression), compressed trie, and **BM25** ranking; integrated Hutter Prize techniques (**cmix/fx2** context mixing) plus hybrid retrieval with sentence-transformer embeddings and **USearch HNSW**.

WORK EXPERIENCE

- FEB 2024 **Senior Frontend Engineer** - TAXTAKER, AUSTIN, TX (REMOTE)
- NOV 2024 Designed an encrypted **localStorage** library for secure client-side data handling.
- OCT 2022 **Data Visualization Engineer** - SPONTIVLY, ST. PETERSBURG, FL (REMOTE)
- SEP 2023 Built interactive dashboards for enterprise clients including **Balenciaga** and the **NFL**.
- SEP 2021 **Junior Frontend Engineer** - ASTRO MALAYSIA, KUALA LUMPUR (REMOTE)
- OCT 2022 Built an interactive tournament medal table for Commonwealth Games 2022 on stadiumastro.com.
- JUN 2021 **Junior Frontend Developer** - CILANS SYSTEMS, AHMEDABAD (REMOTE)

SKILLS

Math: linear algebra, information theory (Shannon entropy), bayesian inference, probability theory, convex/stochastic optimization

Modeling: transformer architectures (encoder/decoder, sparse attention, KV caching), tokenizer design (BPE), training pipelines (pretraining, fine-tuning (SFT, GRPO), ablation design), mechanistic interpretability

Systems: Python, PyTorch, vLLM, SGLang, Weights & Biases, L^AT_EX

AWARDS

2026 RAISE-26 Finalist — selected as 1 of 17 teams from 188 registrations; applied agent-based computational modeling and society simulation to public informatics challenges at Rutgers University | 2022 SIH Judge | 2021 Microsoft Imagine Cup World Finals Runner-Up | 2020 HackMIT Health Tech Winner | 2019 SAP Analytics Cloud Winner